

Experiment 22

Title

Write a Prolog Program to print whether a particular bird can fly or not.

Aim

To implement a Prolog program to determine whether a particular bird can fly using facts and queries.

Objective

To understand the use of facts and rules in Prolog by identifying whether a given bird can fly or not.

Algorithm

Step 1: Define facts for different birds and their flying ability.

Step 2: Create a rule to check whether a bird can fly.

Step 3: Enter the bird name through a query.

Step 4: Prolog searches the database for the given bird.

Step 5: Display whether the bird can fly or not.

Program

```
% Bird Database
```

```
bird(ostrich, cannot_fly).
```

```
bird(penguin, cannot_fly).
```

```
bird(kiwi, cannot_fly).
```

```
bird(eagle, can_fly).
```

```
bird(pigeon, can_fly).
```

```
bird(sparrow, can_fly).
```

```
% Rule to check flying ability
```

```
can_fly(Bird) :-
```

```
    bird(Bird, can_fly).
```

cannot_fly(Bird) :-

bird(Bird, cannot_fly).

Sample Input

?- can_fly(eagle).

?- can_fly(ostrich).

?- bird(Bird, cannot_fly).

Sample Output

```
| ?- consult('C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp22.pl').  
compiling C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp22.pl for byte code...  
C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp22.pl compiled, 15 lines read - 1131 bytes written, 3 ms  
  
(15 ms) yes  
| ?- can_fly(eagle).  
  
yes  
| ?- can_fly(ostrich).  
  
no  
| ?- bird(Bird, cannot_fly).  
  
Bird = ostrich ?  
  
yes  
| ?-  
bird(Bird, cannot_fly).  
  
Bird = ostrich ? ;  
  
Bird = penguin ? ;  
  
Bird = kiwi ? ;  
  
(78 ms) no  
| ?-  
|
```

Result

Hence, the Prolog program to determine whether a particular bird can fly or not was executed successfully and the correct output was obtained.